



LUNDS
UNIVERSITET

FMN050 VT06: Numerical analysis for E/I
Numerisk Analys, Matematikcentrum

Project 1

Last hand-in date: **7 April 2006 at 13:15**

Aim: To study interpolation in an applied problem; the role the basis functions play for the accuracy (conditioning); why it is important for modelling purposes to keep the degree of the interpolation polynomial low.

Grading: Accounts for 10p of the course's total 50p (5p required for passing grade).

Project reports: *The project must be solved and reported individually.* You are encouraged to discuss the problems with your fellow students, but you must construct your own MATLAB programs yourself, run them, interpret the results, and report on your findings in your own report to be graded.

Notes: Make sure that your report has your name, "personnummer," section and your class year, e.g. E03. In your report you must acknowledge all help you received, e.g. from discussions with fellow students or with an instructor. Finally, you should end your report by summing up what you have learned from solving the problems. *Reports must be handed in printed on paper, with accompanying graphics and figures. No electronic submission can be accepted.*

The projects are part of the examination, and your approved and graded project reports remain valid for one year. Read all instructions on the course web site.

In this computer assignment we are going to study interpolation and least squares approximation. In order to solve an applied problem we need to construct a small MATLAB function,

```
function x = lsq(A,y)
```

that solves an overdetermined linear system $Ax = y$. Note that this is straightforward to do in MATLAB using the `\` operator.

Problem 1. Write the function `lsq` as described above, and test it by verifying that it actually delivers the least squares solution in a problem of your choice. For example, take any system consisting of three equations and two unknowns, so that you can solve it by hand, and verify that you get the same solution in MATLAB. Write in your report what system you chose, and what solution you obtained.

(1p)

Problem 2. In the weekly exercise material, you found the recipe for oatmeal porridge, taken from an “Axa havregryn” package. Below, in Table 1, you find how the number of servings x (volume of oatmeal in dl) is related to the corresponding volume of water y (also in dl).

x	1	2	4
y	2.5	4.5	9

Table 1: Data for “Axa havregrynsgröt.”

2a) Use your program above, to determine how much water is needed for 3 servings, assuming that the amount of water varies linearly with the number of servings, i.e.,

$$y(x) = c_0 + c_1x.$$

In order to solve this problem, construct a MATLAB function

```
function A = vdm(xpoints, degree)
```

that builds the Vandermonde matrix

$$A = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots \\ 1 & x_1 & x_1^2 & \dots \\ 1 & \vdots & \vdots & \\ 1 & \vdots & \vdots & \end{pmatrix}$$

where the highest degree monomial is given by the parameter **degree**, and where the points $x_0, x_1, \dots$ are contained in the vector **xpoints**. Make sure that **xpoints** is a column vector, and generate “powers of x ” by using MATLAB’s “dot” notation. Verify that your programs work by fitting the straight line to the given data, by taking the right-hand side vector y from the table. (2p)

2b) Verify that the program also works to solve the interpolation problem, i.e., use the mathematical model

$$y(x) = c_0 + c_1x + c_2x^2.$$

Find the coefficients c_i , and compute the amount of water for three servings, $y(3)$. (1p)

Note that once you have the coefficients c_i , you can make use of MATLAB’s built-in function **polyval** to evaluate $y(x)$ in both tasks above. Run **help polyval** to figure out how to use that function.

We are now ready to solve some real problems. The following is a selection of problems taking from one of the recommended books for the course, Tim Sauer’s *Numerical Analysis*.

CPU	year	transistors
4004	1971	2,250
8008	1972	2,500
8080	1974	5,000
8086	1978	29,000
286	1982	120,000
386	1985	275,000
486	1989	1,180,000
Pentium	1993	3,100,000
Pentium II	1997	7,500,000
Pentium III	1999	24,000,000
Pentium 4	2000	42,000,000
Itanium	2002	220,000,000
Itanium 2	2003	410,000,000

Table 2: Number of transistors on computer chip

Problem 3. According to Moore’s law, the number of transistors on a computer chip doubles every 18 months. Data are available in Table 2.

If the number doubles in a fixed time interval, the number of transistors *grows exponentially*, i.e.,

$$y = k \cdot e^{c \cdot t}$$

where t represents time. We need to determine the two parameters k and c . Note that

$$\log y = \log k + c \cdot t$$

so the problem can be solved by fitting a straight line (finding $\log k$ and c above) to the logarithm of the data.

Use your programs that you have built to solve this problem. Graph the data in a lin–log diagram (“semilog” plot in MATLAB’s terminology; try `help semilogy`). Determine the parameters k and c . Once you have determined these parameters, determine from your exponential model *in how long time the number of transistors double*. Is Moore’s law correct? Make sure to also plot the best straight line in you graph. (2p)

Problem 4. The “Body mass index” or BMI is calculated as your weight (kg) divided by the square of your height (m). It is often claimed that this should be constant. Of course, over time, if your BMI varies you either put on weight or lose weight if the BMI is not constant, but should the BMI have a similar value for different persons, and in particular, is it of relevance for children who grow up?

We shall assume that the weight W of a child is correlated to his/her height h according to a *power law*, i.e.,

$$W = \alpha h^\beta.$$

If the BMI model were “relevant” for children, then we would have $\beta = 2$, but here the task is to determine the parameters α and β . We use average data for children age 2 – 11 according to Table 3.

age	height	weight
2	0.912	13.7
3	0.986	15.9
4	1.060	18.5
5	1.130	21.3
6	1.190	23.5
7	1.260	27.2
8	1.320	32.7
9	1.380	36.0
10	1.410	38.6
11	1.490	43.7

Table 3: Height and weight in children of various ages.

If the model is a power law, the best thing is usually to take logarithms, to get

$$\log W = \log \alpha + \beta \log h.$$

Thus we need to fit a straight line to $\log W$. Use your program to fit the best straight line and find out what the value of β is according to this model. Graph the weight as a function of height, both with linear scales on the axes, and in a log-log diagram. The graphs must contain both the data points and the computed straight line. (2p)

Problem 5. In climate models it is important to study the progression of the mean atmospheric concentration of carbon dioxide. Here we have the data, in parts per million (ppm) in Table 4.

Year	CO ₂ (ppm)
1800	280
1850	283
1900	291
2000	370

Table 4: Carbon dioxide concentration as a function of time.

Find the interpolation polynomial of degree 3 that interpolates the data, and calculate the carbon dioxide concentration at 1950. Finally try to predict the carbon dioxide concentration in 2050.

What is the condition number of the matrix in your calculations?

Rework this problem by using the basis functions

$$\phi_j(t) = (t - 1900)^j$$

and recalculate the condition number for this new basis. Which one is to be preferred and why? (2p)

Report your findings by describing problem, method and results. Include some of the more important programs or code segments, use graphics to illustrate your results, and comment on your findings.

End your report by answering the question of what you have learned in this assignment. Try to think of everything you learned, whether it's about MATLAB, mathematics or numerics. Do not forget to acknowledge any help you have received.